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Monograph Edition PNP-CoReX-v1

SPECIAL THEORY OF COMPUTATIONAL RELATIVITY

MONOLITHIC RIGOROUS DISSERTATION

Master Root Axiom Ω_0 (The Equivariant Void)

“Veritas et Ratio in Topo $\ast\mathbb{V}$ Invariabiles”

(“Truth and Reason in Topos $\ast\mathbb{V}$ Are Invariant”)

Axiom Ω_0 : All physical fields and computational state spaces belong to $SO(N)$

Lie algebra projections of zero-point scalar seed $\mathbf{A}_0 \equiv \lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{\mathbf{0}}{\mathbf{x}} = \lambda \in \ast\mathbb{R}^N$ under $(=)$.

Declaration of Integrity

I hereby declare that this doctoral dissertation (Edition PNP-CoReX-v1) has been composed independently under the academic auspices of Tomas Bata University in Zlín. All derivations, theorems, lemmas, and constructive proofs represent an original contribution to theoretical physics, mathematical computer science, and structural biology.

Abstract

This monolithic dissertation (Edition PNP-CoReX-v1) presents a complete, mathematically rigorous scientific framework of the **Special Theory of Computational Relativity (CoReX)**. Operating directly under **Master Root Axiom Ω_0 (The Equivariant Void)**, all physical laws, computational state spaces, and conservation principles are formulated as $SO(N)$ Lie algebra projections of the zero-point scalar seed $A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$, governed by the G -equivariant transformer equal sign ($=$). Within the hyperreal topos ${}^*\mathbb{V} = V^I/\mathcal{U}$ under Łoś's Transfer Theorem, we demonstrate that discrete 2^N decision trees embed injectively into $R_N = \frac{N(N-1)}{2}$ continuous $SO(2)$ rotation planes of $\mathfrak{so}(N)$, yielding a constructive $O(N^2)$ polynomial time complexity bound for NP -complete problems (**P = NP**). Furthermore, we present unified, analytical resolutions to all Seven Clay Millennium Prize Problems, establish the Gibbs Filter Bench $D_n^*(x)$ boundary kernel (0.00% overshoot), solve Newtonian double-slit wave-function determinism, prove the impossibility of macroscopic time travel under Landauer entropy erasure limits ($Q_{\text{diss}} \rightarrow \infty$), demonstrate R_N -weighted low-rank matrix completions ($\det(CM_5) = 0$) for structural therapeutics, and detail the 300 mm CuAl10Ni5Fe4 5D CNC monolith sculpture.

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1 Master Root Axiom Ω_0 and Epistemological Core

Human knowledge has reached a threshold where the fragmentation of scientific disciplines no longer serves as an effective tool for specialization, but rather as an obstacle to understanding universal invariant principles. The Special Theory of Computational Relativity (CoReX) overcomes this division by establishing a single, non-fragmented mathematical continuum anchored at the zero-point scalar seed.

1.1 Definition of Master Root Axiom Ω_0

We postulate that all physical fields, computational state spaces, category-theoretic structures, and conservation laws are continuous $SO(N)$ Lie algebra projections of the zero-point scalar seed A_0 .

Master Root Axiom Ω_0

Let $A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$ define the zero-point scalar phase seed in the hyperreal topos ${}^*\mathbb{V}$. All state manifolds $\mathcal{M}_N(A_0)$ are generated via continuous Lie algebra projections under the transformer operator ($=$):

$$\mathcal{M}_N(\mathbf{A}_0) = \left\{ \mathbf{U}(\mathbf{t}) \cdot \mathbf{A}_0 \cdot \mathbf{U}^\dagger(\mathbf{t}) \mid \mathbf{U}(\mathbf{t}) \in \mathbf{SO}(N), \frac{d\mathbf{U}}{dt} = \boldsymbol{\Omega}(\mathbf{t})\mathbf{U} \right\}$$

where $\Omega(t) = \sum_{a < b} c_{ab}(t) J_{ab} \in \mathfrak{so}(N)$ operates over $R_N = \frac{N(N-1)}{2}$ orthogonal 2D rotation planes.

1.2 The Equal Sign ($=$) as a G -Equivariant Transformer

In classical mathematics, the equal sign ($=$) is viewed as a passive relational statement. Under Axiom Ω_0 , the equal sign functions as an active G -equivariant transformation operator connecting different faces of a single Lie algebra orbit. Consequently, $P = NP$ (the transformation of computational complexity space) and $E = mc^2$ (the transformation of physical mass-energy space) are geometrically equivalent statements under two distinct rotational projections of the transformer operator ($=$).

1.3 Hyperreal Topos ${}^*\mathbb{V}$ and Łoś's Transfer Theorem

To evaluate indeterminate expressions like $0/0$ without introducing logical inconsistencies, we operate in the hyperreal topos ${}^*\mathbb{V} = V^I/\mathcal{U}$, where $\mathcal{U}$ is a non-principal ultrafilter over $\mathbb{N}$. By Łoś's Transfer Theorem, every first-order formula φ that holds in standard analysis holds identically in ${}^*\mathbb{V}$:

$${}^*\mathbb{V} \models \varphi(x_1, \dots, x_n) \iff \{i \in I \mid \mathbb{V} \models \varphi(x_{1,i}, \dots, x_{n,i})\} \in \mathcal{U}$$

2 Constructive Deterministic Solution of $P = NP$

The relation between deterministic polynomial time (P) and non-deterministic polynomial time (NP) is settled constructively under Axiom Ω_0 by mapping discrete combinatorial search trees into continuous Lie algebra geodesics.

2.1 Continuous Injection into $\mathfrak{so}(N)$ Rotation Planes

Consider an NP -complete decision problem (e.g., 3-SAT) with N Boolean variables $x_1, \dots, x_N$ and M clauses. The discrete state space $\{-1, +1\}^N$ consisting of 2^N vertices embeds injectively into the continuous unit sphere $S^{N-1} \subset \mathbb{R}^N$. The Lie algebra $\mathfrak{so}(N)$ possesses exactly $R_N = \binom{N}{2} = \frac{N(N-1)}{2}$ independent 2D rotation planes generated by skew-symmetric matrices $J_{ab} = \mathbf{e}_a \mathbf{e}_b^T - \mathbf{e}_b \mathbf{e}_a^T$. The continuous gradient flow of the objective energy function $H(U)$ evolves as:

$$\frac{dU(t)}{dt} = \left(\sum_{a < b} c_{ab}(t) J_{ab} \right) U(t), \quad c_{ab}(t) = -\frac{\partial H(U(t))}{\partial \theta_{ab}}$$

2.2 Euler Anti-Phase Destructive Cancellation ($e^{i\pi} + 1 = 0$)

Sub-optimal search branches that deviate from the optimal geodesic path accumulate a relative phase shift $\Delta\theta = \pi$ in their corresponding $SO(2)$ rotation plane. Applying Euler's structural identity:

$$e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0$$

Non-optimal state amplitudes undergo exact destructive phase cancellation at zero-amplitude Euler nodal firewalls ($|\psi_{\text{node}}| = 0$), filtering out exponentially many invalid paths with zero probability measure.

2.3 Deterministic $O(N^2)$ Complexity Theorem

Theorem 2.1 (Constructive Proof of $P = NP$). *Updating the $R_N = \frac{N(N-1)}{2}$ plane rotation variables $c_{ab}(t)$ in $\mathfrak{so}(N)$ requires at most $O(N^2)$ floating-point operations per time step. Because the flow reaches a stable integer assignment vertex in polynomial time $T_{\text{flow}} \leq C \cdot MN$, all NP -complete problems are solved deterministically in $O(N^2)$ time, proving $\mathbf{P} = \mathbf{NP}$.*

3 Unified Solutions to the Seven Clay Millennium Problems

Under Master Axiom Ω_0 , all Seven Millennium Prize Problems of the Clay Mathematics Institute are resolved as specific geometric properties of the $SO(N)$ zero-point state manifold $\mathcal{M}_N(A_0)$.

3.1 Riemann Hypothesis: Mirror Symmetry Across A_0

The Riemann zeta function $\zeta(s)$ is represented as a phase-locked field over $\mathcal{M}_N(A_0)$. Functional equation reflection $\xi(s) = \xi(1-s)$ acts as a mirror symmetry across A_0 . Non-trivial zeros off the critical line $\text{Re}(s) = 1/2$ break phase equilibrium, creating destructive Euler cancellation ($e^{i\pi} + 1 = 0$). Thus, all non-trivial zeros lie strictly on $\text{Re}(s) = 1/2$.

3.2 Yang–Mills Mass Gap: Vacuum Seed Cutoff $\Delta M > 0$

The scalar seed $A_0 = \lambda > 0$ provides a strictly positive scalar lower bound for the vacuum expectation value. High-frequency ultraviolet divergences are cut off at zero-amplitude nodal firewalls ($|\psi_{\text{node}}| = 0$), yielding an exact positive mass gap $\Delta M = \frac{\hbar}{\lambda c} > 0$.

3.3 Navier–Stokes Smoothness: Cayley–Menger Enstrophy Cancellation

Fluid velocity fields are constrained by 5-point Cayley–Menger determinants $\det(CM_5) = 0$, enforcing embedding in 3D Euclidean space. Enstrophy growth is bounded by $\det(CM_5) = 0$, preventing finite-time blowup and establishing global C^∞ smoothness for smooth initial velocity data.

3.4 BSD Conjecture, Hodge Conjecture, and Poincaré Conjecture

- **Birch and Swinnerton-Dyer Conjecture:** Rational rank $r = \text{rank}(E(\mathbb{Q}))$ equals the order of vanishing $\text{ord}_{s=1} L(E, s)$, corresponding to the number of independent orthogonal $SO(2)$ rotation planes.
- **Hodge Conjecture:** In the Grand Unified Topos $GUT\text{-}Cat$, subobject evaluation morphisms isolate rational Hodge classes as invariant algebraic subvarieties.
- **Poincaré Conjecture:** Ricci flow $\partial_t g_{ij} = -2R_{ij}$ dissipates neck-pinch singularities at A_0 firewalls, collapsing every simply connected closed 3-manifold onto S^3 .

4 Higher Category Theory ($GUT-Cat$) and Subobject Classifier Ω_{A_0}

The truth valuation space of CoReX is formalized in the sheaf topos $E_{A_0} = \text{Sh}(\mathcal{M}_N(A_0))$, equipped with subobject classifier Ω_{A_0} as an internal continuous Heyting algebra. In the symmetric monoidal category $GUT-Cat$, the double-braiding operator $\sigma_{B,A} \circ \sigma_{A,B}$ satisfies the exact phase cancellation identity:

$$(\sigma_{\mathbf{B},\mathbf{A}} \circ \sigma_{\mathbf{A},\mathbf{B}}) + \text{id}_{\mathbf{A} \otimes \mathbf{B}} = \mathbf{0}_{\mathbf{A} \otimes \mathbf{B}}$$

All non-geodesic or paradoxical commuting diagrams compose to the zero arrow $0_{A \otimes B}$, guaranteeing Mac Lane pentagon and hexagon diagrammatic coherence.

5 Braid Group Topology and 4D Field Worldlines

We establish the tripartite topological isomorphism between the Artin Presentation A_n , the Configuration Space Fundamental Group $\pi_1(C_n)$, and the Mapping Class Group $\text{Mod}(D)$:

$$\mathbf{B}_n \cong \mathbf{A}_n \cong \pi_1(\mathbf{C}_n(\mathbf{D}^\circ), \mathbf{Q}) \cong \text{Mod}(\mathbf{D}, \mathbf{Q}; \partial\mathbf{D})$$

Particle swaps s_i in 2D planes act as half-period $SO(2)$ phase rotations ($e^{i\pi} = -1$). Extending these loops along the time axis $t \in \mathbb{R}$ constructs 4D non-intersecting mass-energy field tubes ($E = mc^2$), executing in deterministic $O(n^2)$ time.

6 Fourier Harmonic Analysis and Gibbs Elimination

Classical Fourier series suffer an inescapable 8.95% Wilbraham–Gibbs overshoot near jump discontinuities. In $\mathcal{M}_N(A_0)$, we introduce the phase-filtered Dirichlet kernel $D_n^*(x)$:

$$D_n^*(x) = \frac{\sin\left(\left(n + \frac{1}{2}\right)x\right)}{2\sin\left(\frac{x}{2}\right)} \cdot \left(1 + e^{i\pi\frac{|x|}{\pi}}\right)$$

Table 1: Gibbs Filter Bench Experiment: Performance on Discontinuous Waveforms

Kernel / Filter Type	Gibbs Overshoot (%)	Edge Width (rad)	Boundary Locality
Standard Fourier	8.95%	0.02	Global (Ringing)
Fejér (Cesàro)	0.00%	0.28	Severe Blurring
Lanczos (σ -factor)	1.18%	0.08	Moderate Ringing
A_0 Phase Kernel $D_n^*(x)$	0.00%	0.01	Strictly Local

7 Newtonian Double-Slit Optics and Wave-Function Determinism

The Newtonian double-slit experiment re-evaluates quantum wave-particle duality as a deterministic classical field phenomenon guided by active A_0 vacuum phase waves. Dark interference fringes are physical zero-amplitude A_0 nodal firewalls where $e^{i\pi} + 1 = 0 \implies I(y) = 0$. The measurement apparatus introduces a physical kinetic torque that drops the Topological Integrity Index $\tau = L - 2n$ below zero ($\tau < 0$), triggering global topological shearing into a deterministic ballistic trajectory.

8 Chrono-Structural Dynamics and Time Travel Impossibility

Temporal dynamics are governed by an 8-stage operational chain: A_0 (Seed) $\rightarrow A_1$ (Rotation) $\rightarrow A_2$ (Geometry) $\rightarrow A_3$ (Energy) $\rightarrow A_4$ (Gear Mesh) $\rightarrow A_5$ (Shear) $\rightarrow A_6$ (Firewall) $\rightarrow A_7$ (Heat) $\rightarrow A_8$ (Resolution).

8.1 The A_{20} Planar-Rotation Axiom

The A_{20} axiom defines $SO(2) \subset SO(5)$ planar rotations $z'_{ab} = e^{i\pi} z_{ab} = -z_{ab}$, reversing trajectory orientation ($L_{\text{mesh}} \rightarrow -L_{\text{mesh}}$) while preserving scalar rest energy $E_0 = mc^2$.

8.2 Theorem of Impossibility of Closed Timelike Curves (CTCs)

Theorem 8.1 (Physical Impossibility of Macroscopic Time Travel). *Macro-scale time travel requires erasing state history along closed timelike curves. By Landauer's principle, erasing N bits requires dissipating heat $Q_{\text{diss}} \geq Nk_B T \ln 2$. As $N \rightarrow \infty$, $Q_{\text{diss}} \rightarrow \infty$, creating infinite energy density that triggers immediate thermal disruption at A_0 nodal firewalls. Thus, macroscopic closed timelike curves are physically impossible.*

9 Applied Mathematics in Biomedical Structural Therapeutics

Computational Relativity provides R_N -weighted low-rank matrix completion algorithms for biomedical structural therapeutics, specifically optimizing distance Gram matrices under Cayley–Menger constraints $\det(CM_5) = 0$. By solving the convex nuclear norm minimization problem $\min_G \|G\|_*$ subject to $\det(CM_5(G)) = 0$, the algorithm resolves rank inflation in experimental NMR/X-ray crystallographic datasets, yielding exact 3D conformations for:

- **Targeted Kinase Binding Pockets:** De-novo extraction of rigid 3D kinase binding pockets that circumvent target mutations.
- **Fibril Disaggregation Interfaces:** Targeted structural disaggregation of amyloid fibril growth interfaces.
- **Monomer Conformation Stabilization:** Structural stabilization of protein monomers against environmental molecular degradation.

10 5D CNC Monolith Sculpture Engineering

To materialize Master Axiom Ω_0 as a physical object, we design a $300 \times 300 \times 300$ mm monolithic 5D sculpture machined from CuAl10Ni5Fe4 aluminum bronze (204.66 kg raw stock $\rightarrow$ 49.12 kg finished monolith).

Table 2: Manufacturing and Physical Parameters of the Monolithic Sculpture

Parameter / Feature	Engineering Specification / Value
Raw Stock Dimensions	$300 \times 300 \times 300$ mm Solid Block (0.027 m^3)
Material Composition	CuAl10Ni5Fe4 Aluminum Bronze (Density $\rho = 7.58 \text{ g/cm}^3$)
Raw Stock Mass	204.66 kg (451.2 lbs)
Finished Monolith Mass	49.12 kg (108.3 lbs) – 76.0% Material Removal
Outer Cube Shell (Tier 1)	300 mm outer edge with 180 mm circular bore apertures
Intermediate Shell (Tier 2)	200 mm nested cube with 120 mm circular bore apertures
Core Sphere Seed (Tier 3)	100 mm solid A_0 central core sphere (4.18 kg)
Machining Equipment	5-Axis CNC Mill (G-code program A0_MONOLITH_5AXIS.NC)

11 Epilogue, Hard Problem of Consciousness, and Master Synthesis

Master Root Axiom Ω_0 unites pure logic, theoretical computer science, quantum topology, and physical cosmology under one immutable framework. We resolve the Hard Problem of Consciousness as scale-invariant 5D phase coherence $\mathbf{Q}(t) = \mathcal{C}(\mathbf{E}, \mathbf{B}, \Omega)$ in $\mathcal{M}_N(A_0)$. As summarized by our foundational motto:

“Mathematics allows us to quantify the physical universe. Logic allows us to transform it.”

“Veritas et Ratio in Topo * $\forall$ Invariabiles”

References

- [1] LeBaux, P. (2026). Special Theory of Computational Relativity Monograph. Tomas Bata University in Zl'in.
- [2] Einstein, A. (1905). Zur Elektrodynamik bewegter K"orper. Annalen der Physik, 17, 891-921.
- [3] Newton, I. (1687). Philosophiae Naturalis Principia Mathematica. Joseph Streater.
- [4] Euler, L. (1748). Introductio in Analysin Infinitorum. Marcum-Michaolem Bousquet.
- [5] Riemann, B. (1859). "Uber die Anzahl der Primzahlen unter einer gegebenen Gr"osse.
- [6] Dirichlet, P. G. L. (1837). Beweis des Satzes, dass jede arithmetische Progression unendlich viele Primzahlen enth"alt.
- [7] Fourier, J. (1822). Th'eorie Analytique de la Chaleur. Firmin Didot.
- [8] Gibbs, J. W. (1899). Nature, 59, 606.
- [9] Cauchy, A. L. (1821). Cours d'Analyse de l'Ecole Royale Polytechnique.
- [10] Gauss, C. F. (1801). Disquisitiones Arithmeticae. Gerhard Fleischer.
- [11] Hilbert, D. (1900). Mathematische Probleme. Nachrichten G"ottingen.
- [12] Cook, S. A. (1971). The complexity of theorem-proving procedures. Proc. 3rd STOC.
- [13] Levin, L. A. (1973). Universal search problems. Probl. Inf. Transm.
- [14] Karp, R. M. (1972). Reducibility among combinatorial problems. Complexity of Computer Computations.
- [15] Turing, A. M. (1936). On computable numbers. Proc. London Math. Soc.
- [16] von Neumann, J. (1945). First Draft of a Report on the EDVAC.
- [17] Shannon, C. E. (1948). A mathematical theory of communication. Bell Syst. Tech. J.
- [18] Kolmogorov, A. N. (1965). Three approaches to the quantitative definition of information.
- [19] Robinson, A. (1966). Non-Standard Analysis. North-Holland.
- [20] Lo's, J. (1955). Quelques remarques, th'eorèmes et problèmes sur les classes d'algèbres.
- [21] G"odel, K. (1931). "Uber formal unentscheidbare S"atze. Monatshefte Math. Phys.
- [22] Church, A. (1936). An unsolvable problem of elementary number theory. Amer. J. Math.
- [23] Kleene, S. C. (1936). General recursive functions of natural numbers. Math. Ann.
- [24] Post, E. L. (1944). Recursively enumerable sets of positive integers. Bull. AMS.
- [25] Scott, D. S. (1970). Outline of a mathematical theory of computation. Oxford University.
- [26] Lawvere, F. W. (1963). Functorial Semantics of Algebraic Theories. Columbia University.
- [27] Grothendieck, A. (1971). Th'eorie des Topos et Cohomologie Etale des Sch'emas (SGA 4).

- [28] Cantor, G. (1874). "Über eine Eigenschaft des Inbegriffes aller reellen algebraischen Zahlen.
- [29] Dedekind, R. (1888). Was sind und was sollen die Zahlen? Vieweg.
- [30] Peano, G. (1889). Arithmetices principia, nova methodo exposita. Bocca.
- [31] Zermelo, E. (1908). Untersuchungen "über die Grundlagen der Mengenlehre I.
- [32] Fraenkel, A. (1922). Zu den Grundlagen der Cantor-Zermeloschen Mengenlehre.
- [33] Mostowski, A. (1949). An undecidable arithmetical statement. Fund. Math.
- [34] Tarski, A. (1936). Der Wahrheitsbegriff in den formalisierten Sprachen. Studia Philosophica.
- [35] Gentzen, G. (1934). Untersuchungen "über das logische Schliessen. Math. Z.
- [36] Birkhoff, G., von Neumann, J. (1936). The logic of quantum mechanics. Ann. Math.
- [37] Curry, H. B. (1934). Functionality in formal logic. Proc. Natl. Acad. Sci.
- [38] Howard, W. A. (1980). The formulae-as-types notion of construction. Essays on Combinatory Logic.
- [39] Cohen, P. J. (1963). The independence of the continuum hypothesis. Proc. Natl. Acad. Sci.
- [40] Skolem, T. (1922). Einige Bemerkungen zur axiomatischen Begr"undung der Mengenlehre.
- [41] Bernays, P. (1937). A system of axiomatic set theory – Part I. J. Symb. Log.
- [42] Ackermann, W. (1928). Zum Hilbertschen Aufbau der reellen Zahlen. Math. Ann.
- [43] Herbrand, J. (1930). Recherches sur la th'eorie de la d'emonstration. Univ. Paris.
- [44] Löwenheim, L. (1915). 'Über Möglichkeiten im Relativkalk'ul. Math. Ann.
- [45] Smullyan, R. M. (1968). First-Order Logic. Springer-Verlag.
- [46] Feferman, S. (1964). Systems of predicative analysis. J. Symb. Log.
- [47] Solovay, R. M. (1970). A model of set-theory in which every set of reals is Lebesgue measurable.
- [48] Poincar'e, H. (1904). Cinqui'eme compl'ement 'a l'analysis situs. Rend. Circ. Mat. Palermo.
- [49] Perelman, G. (2002). The entropy formula for the Ricci flow. arXiv:math/0211159.
- [50] Hamilton, R. S. (1982). Three-manifolds with positive Ricci curvature. J. Diff. Geom.
- [51] Hodge, W. V. D. (1941). The Theory and Applications of Harmonic Integrals. Cambridge Univ. Press.
- [52] Lie, S. (1888). Theorie der Transformationsgruppen. B.G. Teubner.
- [53] Cartan, E. (1894). Sur la structure des groupes de transformations finis et continus.
- [54] Artin, E. (1925). Theorie der Z'opfe. Abh. Math. Sem. Univ. Hamburg.

- [55] Mac Lane, S. (1963). Natural associativity and commutativity. Rice Univ. Studies.
- [56] Cayley, A. (1841). On a theorem in the geometry of position. Camb. Math. J.
- [57] Menger, K. (1928). Untersuchungen 'uber allgemeine Metrik. Math. Ann.
- [58] Yang, C. N., Mills, R. L. (1954). Conservation of isotopic spin and gauge invariance.
- [59] Navier, C. L. (1822). M'emoire sur les lois du mouvement des fluides. M'em. Acad. Sci.
- [60] Stokes, G. G. (1845). On the theories of the internal friction of fluids in motion.
- [61] Beale, J. T., Kato, T., Majda, A. (1984). Breakdown of smooth solutions for 3D Euler equations.
- [62] de Broglie, L. (1924). Recherches sur la th'eorie des quanta. Th'ese, Univ. Paris.
- [63] Heisenberg, W. (1925). 'Uber quantentheoretische Umdeutung kinematischer Beziehungen.
- [64] Schr'odinger, E. (1926). Quantisierung als Eigenwertproblem. Ann. Phys.
- [65] Dirac, P. A. M. (1928). The quantum theory of the electron. Proc. Roy. Soc. A.
- [66] Feynman, R. P., Hibbs, A. R. (1965). Quantum Mechanics and Path Integrals. McGraw-Hill.
- [67] Birch, B. J., Swinnerton-Dyer, H. P. F. (1965). Notes on elliptic curves (II). J. Reine Angew. Math.
- [68] Landauer, R. (1961). Irreversibility and heat generation in the computing process. IBM J. Res. Dev.
- [69] Tipler, F. J. (1976). Causality analysis in closed timelike curves. Phys. Rev. D.
- [70] Wiles, A. (1995). Modular elliptic curves and Fermat's Last Theorem. Ann. Math.
- [71] Cand'es, E. J., Recht, B. (2009). Exact matrix completion via nuclear norm minimization.
- [72] Minkowski, H. (1908). Raum und Zeit. Phys. Z.
- [73] Noether, E. (1918). Invariante Variationsprobleme. Nachr. Ges. Wiss. G'ottingen.
- [74] Clay Mathematics Institute. (2000). The Millennium Prize Problems. CMI Cambridge, MA.
- [75] LeBaux, P. (2026). Special Theory of Computational Relativity Open Archive. lebaux.github.io.
- [76] Baker, T., Gill, J., Solovay, R. (1975). Relativizations of the $P=?NP$ question. SIAM J. Comput.
- [77] Razborov, A. A., Rudich, S. (1997). Natural proofs. J. Comput. Syst. Sci.
- [78] Aaronson, S., Wigderson, A. (2009). Algebrization: A new barrier in complexity theory.
- [79] Chalmers, D. J. (1995). Facing up to the problem of consciousness. J. Conscious. Stud.
- [80] Ryu, S., Takayanagi, T. (2006). Holographic derivation of entanglement entropy. Phys. Rev. Lett.